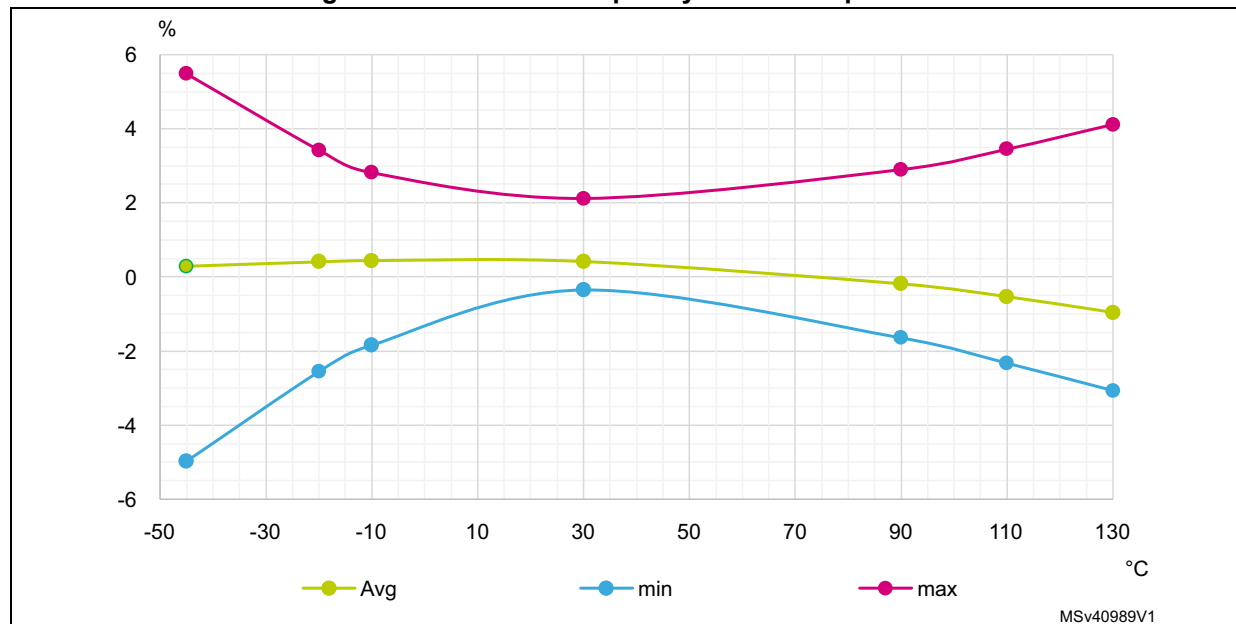


Table 42. HSIUSB48 oscillator characteristics (continued)

Symbol	Parameter	Conditions ⁽¹⁾	Min	Typ	Max	Unit
NT jitter	Next transition jitter Accumulated jitter on 28 cycles ⁽⁴⁾	-	-	$\pm 0.15^{(2)}$	-	ns
PT jitter	Paired transition jitter Accumulated jitter on 56 cycles ⁽⁴⁾	-	-	$\pm 0.25^{(2)}$	-	ns

1. $V_{DD} = 3.0\text{ V}$, $T_A = -40$ to $125\text{ }^{\circ}\text{C}$, unless otherwise specified.
2. Specified by design. Not tested in production.
3. Evaluated by characterization. Not tested in production.
4. Jitter measurement is performed without clock source activated in parallel.

Figure 22. HSIUSB48 frequency versus temperature



Low-speed internal (LSI) RC oscillator

Table 43. LSI oscillator characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{LSI}	LSI frequency	$V_{DD} = 3.3\text{ V}$, $T_A = 25\text{ }^{\circ}\text{C}$	31.04	32	32.96	kHz
		$V_{DD} = 2\text{ V to }3.6\text{ V}$, $T_A = -40\text{ to }125\text{ }^{\circ}\text{C}$	29.5 ⁽¹⁾	-	34 ⁽¹⁾	
$t_{SU(LSI)}^{(2)}$	LSI oscillator start-up time	-	-	80	130	μs
$t_{STAB(LSI)}^{(2)}$	LSI oscillator stabilization time	5% of final frequency	-	125	180	μs
$I_{DD(LSI)}^{(2)}$	LSI oscillator power consumption	-	-	110	180	nA

1. Evaluated by characterization. Not tested in production.
2. Specified by design. Not tested in production.